IC-Knock-Poly: Iterative Conditional Knockoffs for High-Dimensional Sparse Rational Polynomial Regression

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Abstract

We introduce *IC-Knock-Poly*, an algorithm for discovering sparse rational polynomial governing equations from observational data under rigorous false discovery rate (FDR) control. The method integrates three components: (i) a penalised Gaussian mixture model (GMM) with GraphLasso precision matrices that learns the joint distribution of covariates from unlabelled data; (ii) a conditional knockoff generator that exploits the estimated mixture structure to produce valid exchangeable decoys; and (iii) an iterative knockoff–Lasso screening procedure with a *post-selection inference* (PoSI) α-spending budget that controls the *global* FDR across all expansion steps.

A polynomial dictionary Φ maps base features and their knockoffs into monomials of bounded degree, enabling the recovery of interpretable equations such as $y = x_1^2 + 1/x_2$ while preserving exchangeability throughout the expanded space. We derive a non-asymptotic FDR bound and a minimax-rate parameter-estimation guarantee. Simulation studies across a range of sample sizes ($n \in \{100, \ldots, 500\}$), sparsity levels ($k \in \{2, 3, 4\}$), and polynomial degrees ($d \in \{2, 3\}$) confirm that IC-Knock-Poly achieves FDR $= 0$ in all scenarios. Current experiments reveal that the method's GMM-based knockoff generator is conservative: while FDR control is exact, true-positive recall is zero under the default configuration, indicating that the knockoff importance statistics are not yet sufficiently powered to cross the selection threshold. We analyse this behaviour, compare it against four competitive baselines—PolyLasso, PolyOMP, SparsePolySTLSQ, and PolyKnockoff—and discuss concrete paths toward calibrating conservatism.

1 Introduction

Discovering interpretable, parsimonious models from high-dimensional data is a central challenge in scientific machine learning. Many physical, biological, and engineering systems obey governing equations that are *polynomial* or *rational polynomial* in their state variables—Newton's laws, chemical kinetics, and ecological Lotka–Volterra dynamics are canonical examples. The *sparse identification of nonlinear dynamics* (SINDy) framework [4, 15] demonstrated that such equations can be recovered from time-series data by regressing observations onto a "library" of candidate functions. However, SINDy and its extensions treat variable selection as a regularised regression problem without providing statistical guarantees on false discoveries.

In the classical linear setting, the Lasso [17] achieves model selection consistency under restrictive *irrepresentable* conditions [19], and even then controls only marginal inclusion probabilities rather than the FDR. Stability selection [13] and sure independence screening [9] offer different partial remedies but similarly lack finite-sample FDR guarantees in the non-linear case.

The *knockoff* filter [1] provides exact finite-sample FDR control for linear models with known design matrices. *Model-X knockoffs* [6] extended this to settings where only the feature distribution is known, decoupling the knockoff construction from the regression model. Subsequent work introduced deep generative approaches to knockoff sampling [14, 12], but these are designed for linear or generalised linear responses and do not naturally accommodate the polynomial structure required for equation discovery.

Contributions. This paper makes the following contributions:

C1. We formulate *Sparse Rational Polynomial Regression* as a simultaneous basis-expansion and FDR-controlled variable-selection problem (Section 3).

C2. We propose IC-Knock-Poly, the first algorithm to combine penalised GMM distribution learning, conditional knockoff generation, and iterative PoSI α-spending to solve this problem (Section 4).

C3. We prove a non-asymptotic FDR bound (Theorem 1) and a minimax-optimal parameter-estimation guarantee (Theorem 2), together with complete proofs (Section 5).

C4. We provide an open-source Python implementation with CPU-parallel and Rust-accelerated kernels, and a thorough complexity analysis (Section 6).

C5. We report a comprehensive simulation study comparing IC-Knock-Poly with PolyLasso, PolyOMP, SparsePolySTLSQ, PolyCLIME, and PolyKnockoff, including an honest assessment of the current method's conservatism (Sections 7–8).

Paper organisation. Section 2 surveys related work. Section 3 formalises the problem. Section 4 presents the algorithm. Section 5 contains theoretical guarantees and proofs. Section 6 analyses computational complexity. Section 7 reports experiments. Section 8 discusses limitations and future work. Section 9 concludes.

2 Related Work

Sparse regression and model selection. The Lasso [17] minimises the sum of squared residuals plus an ℓ_1 penalty and performs simultaneous estimation and variable selection. Under the restricted isometry property [8] or restricted eigenvalue conditions [5], the Lasso recovers the support consistently when the signal is sufficiently strong relative to noise. The elastic net [21] adds an ℓ_2 penalty to handle correlated predictors. Graphical-model-based methods such as the graphical Lasso [10] estimate sparse precision matrices and underpin our GMM phase. For post-selection inference—making valid inferential statements about coefficients chosen by a data-adaptive procedure—see [3].

SINDy and equation discovery. SINDy [4] writes the derivative of a state vector as a linear combination of candidate library functions and applies sequentially thresholded least squares (STLSQ) to obtain a sparse model. Extensions to partial differential equations [15] and the SR3 relaxation [20] have further broadened applicability. None of these methods controls the FDR, which motivates our work. Our SparsePolySTLSQ baseline implements the STLSQ strategy adapted to observational (non-time-series) polynomial regression.

Knockoffs for variable selection. The fixed-X knockoff filter [1] proved that exchangeable knockoff copies of the design matrix, combined with antisymmetric importance statistics, control the FDR for linear models. Model-X knockoffs [6] lifted this to arbitrary response models given a known feature distribution. Deep knockoffs [14] learn the knockoff generator from data using variational autoencoders. Our conditional knockoff generator (Section 4.3.1) differs from these in that it exploits a Gaussian mixture model for the feature distribution, providing closed-form conditional sampling with GraphLasso precision matrices.

FDR control in multiple testing. The Benjamini–Hochberg procedure [2] controls the FDR for independent tests; [16] introduced the q-value framework. Stability selection [13] controls the expected number of false positives for regularisation paths. Our PoSI α-spending sequence is inspired by the Knockoff+ modification of [1], which raises the threshold to control the *modified* FDP rather than the classical FDP.

Dictionary learning and compressed sensing. Compressed sensing [8] and basis pursuit [7] establish exact recovery of sparse signals from random projections. Polynomial dictionary expansion is a deterministic analogue: we expand p base features into $\tilde{p} = \binom{p+d}{d}$ monomials and seek a k-sparse vector in this expanded space. High-dimensional statistics references [5, 11, 18] provide the technical background for our theoretical analysis.

3 Problem Formulation

3.1 Data model

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ denote an observed feature matrix with rows $\mathbf{x}_i \overset{\text{iid}}{\sim} P_X$, where P_X is the unknown joint distribution of p covariates. Let $\mathbf{y} = (y_1, \ldots, y_n)^\top$ be real-valued responses. We assume the data-generating mechanism is

$$y_i \;=\; f^*(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{1}$$

where $f^* : \mathbb{R}^p \to \mathbb{R}$ belongs to the class of *sparse rational polynomials.*

Definition 1 (Rational polynomial)**.** *A* degree-d rational polynomial *in p variables is any finite sum of monomials $x_1^{e_1} \cdots x_p^{e_p}$ with $e_j \in \{-d, \ldots, d\}$ for all j, excluding the zero exponent. The* support *of such a function is the set of base variables j for which at least one monomial has $e_j \neq 0$.*

3.2 Polynomial dictionary

Fix a maximum exponent $d \geq 1$. For a single base feature x_j define the exponent set $\mathcal{E} = \{-d, \ldots, -1, 1, \ldots, d\}$ of cardinality $m = 2d$. The expanded feature for exponent e is $\phi_e(x_j) = x_j^e$ (with the convention $\phi_e(x_j) = 0$ when $x_j = 0$ and $e < 0$). The full *polynomial dictionary* for column j is

$$\Phi_j(\mathbf{x}_j) \;=\; \big[\phi_{-d}(\mathbf{x}_j),\ \ldots,\ \phi_{-1}(\mathbf{x}_j),\ \phi_1(\mathbf{x}_j),\ \ldots,\ \phi_d(\mathbf{x}_j)\big] \in \mathbb{R}^{n\times m}.$$

Stacking all p columns gives the expanded matrix $\Phi(\mathbf{X}) \in \mathbb{R}^{n\times\tilde{p}}$ with $\tilde{p} = p \cdot m = 2pd$. The response model is then

$$\mathbf{y} \;=\; \Phi(\mathbf{X})\,\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}, \quad \boldsymbol{\beta}^* \in \mathbb{R}^{\tilde{p}}, \tag{2}$$

where $\boldsymbol{\beta}^*$ is s-sparse ($s \ll \tilde{p}$) with support $S^* \subseteq [\tilde{p}]$.

3.3 Goal and FDR criterion

A variable selection procedure outputs an estimated support $\hat{S} \subseteq [\tilde{p}]$. Define

$$\mathrm{FDP}(\hat{S}) \;=\; \frac{\left|\hat{S} \setminus S^*\right|}{\left|\hat{S}\right| \vee 1}, \qquad \mathrm{FDR} \;=\; \mathbb{E}[\mathrm{FDP}(\hat{S})].$$

Our goal is to recover S^* (or a meaningful subset) while maintaining $\mathrm{FDR} \leq Q$ for a user-specified level $Q \in (0, 1)$. Additionally, we require a coefficient estimator $\hat{\boldsymbol{\beta}}$ satisfying a minimax-rate estimation bound.

4 Methodology

IC-Knock-Poly operates in three phases executed in sequence: Phase 1 (distribution learning), Phase 2 (initialisation), and Phase 3 (iterative expansion and screening). Algorithm 1 gives the complete pseudocode.

4.1 Phase 1: Distribution learning via penalised GMM

Because Model-X knockoffs [6] require knowledge of P_X, we estimate it using a penalised GMM. Specifically, we assume

$$P_X \;=\; \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Theta}_k^{-1}), \tag{3}$$

where K is the number of mixture components, $\pi_k > 0$ are mixture weights with $\sum_k \pi_k = 1$, and each precision matrix $\boldsymbol{\Theta}_k$ is estimated with a GraphLasso penalty λ:

$$\hat{\boldsymbol{\Theta}}_k \;=\; \underset{\Theta \succ 0}{\operatorname{argmax}} \left\{ \log\det\Theta - \operatorname{tr}(\hat{\boldsymbol{\Sigma}}_k\,\Theta) - \lambda \left\|\Theta\right\|_1 \right\}, \tag{4}$$

where $\hat{\boldsymbol{\Sigma}}_k$ is the sample covariance for component k. The penalty encourages sparsity in the conditional independence graph of $P_{X\,|\,z=k}$. When λ is not supplied by the user, it is selected by the Bayesian Information Criterion over a grid.

The EM algorithm alternates between soft-assigning observations to components and solving (4) for each component. Fitting uses only available observations (labelled and unlabelled), making the method naturally semi-supervised.

4.2 Phase 2: Initialisation

Set the active polynomial index set $\mathcal{A}^{(0)} = \emptyset$, the active base set $\mathcal{B}^{(0)} = \emptyset$, and the residuals $\mathbf{R}^{(0)} = \mathbf{y}$. Set the iteration counter $t = 0$ and the α-budget remainder $Q_{\text{rem}} = Q$.

4.3 Phase 3: Iterative expansion and screening

4.3.1 Conditional knockoff generation

At iteration t let $\mathcal{B}^{(t)} \subseteq [p]$ be the base features already selected. For each unselected base feature $j \notin \mathcal{B}^{(t)}$ we generate a knockoff $\tilde{X}_j$ that is conditionally exchangeable with X_j given all other features.

Under the mixture model (3), the conditional distribution of X_j given $\mathbf{X}_{-j}$ within component k is

$$P(X_j \,|\, \mathbf{X}_{-j}, z = k) \;=\; \mathcal{N}\Big(\mu_{k,j} - (\boldsymbol{\Theta}_{k,jj})^{-1}\,\boldsymbol{\Theta}_{k,j,-j}\,(\mathbf{X}_{-j} - \boldsymbol{\mu}_{k,-j}),\; (\boldsymbol{\Theta}_{k,jj})^{-1}\Big). \tag{5}$$

We draw $\tilde{X}_j^{(i)}$ by: (a) computing the posterior component probability $\hat{\gamma}_{ik} = P(z_i = k \,|\, \mathbf{x}_i)$ from the estimated GMM; (b) sampling $z_i^* \sim \text{Categorical}(\hat{\gamma}_{i1}, \ldots, \hat{\gamma}_{iK})$; (c) sampling $\tilde{X}_j^{(i)}$ from (5) with $k = z_i^*$.

4.3.2 Expanded dictionary and importance statistics

Let $B = [p] \setminus \mathcal{B}^{(t)}$ be the remaining base features. Form the sub-matrices $\mathbf{X}_B$ and $\tilde{\mathbf{X}}_B$ (knockoffs) and expand both through the polynomial dictionary:

$$\Phi_B \;=\; \big[\Phi(\mathbf{X}_B),\; \Phi(\tilde{\mathbf{X}}_B)\big] \;\in\; \mathbb{R}^{n \times 2|B|m}.$$

Fit a cross-validated Lasso on Φ_B against the current residuals $\mathbf{R}^{(t-1)}$, obtaining coefficients $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_{\text{feat}}, \hat{\boldsymbol{\beta}}_{\text{knock}})$. For each unselected term $(j, e) \in B \times \mathcal{E}$ define the importance statistic

$$W_{(j,e)} \;=\; \left|\hat{\beta}_{(j,e)}\right| - \left|\hat{\beta}^{\text{knock}}_{(j,e)}\right|. \tag{6}$$

4.3.3 PoSI threshold with α-spending

At iteration t allocate budget

$$q_t \;=\; \frac{Q}{t(t+1)}, \tag{7}$$

so that $\sum_{t=1}^{\infty} q_t = Q$. Compute the *Knockoff+* threshold

$$\tau_t \;=\; \min\left\{\omega > 0 \;:\; \frac{1 + \#\{(j,e) : W_{(j,e)} \le -\omega\}}{\#\{(j,e) : W_{(j,e)} \ge \omega\} \vee 1} \le q_t\right\}. \tag{8}$$

Select all terms (j,e) with $W_{(j,e)} \ge \tau_t$.

4.3.4 Support update and stopping rule

Add newly selected polynomial terms to $\mathcal{A}^{(t)}$ and their base indices to $\mathcal{B}^{(t)}$. Refit ordinary least squares (OLS) on $\Phi(\mathbf{X})_{\mathcal{A}^{(t)}}$ and update residuals $\mathbf{R}^{(t)} = \mathbf{y} - \Phi(\mathbf{X})_{\mathcal{A}^{(t)}}\hat{\boldsymbol{\beta}}^{(t)}$. Stop if no new terms are selected or if the budget is exhausted.

Algorithm 1 IC-Knock-Poly

Require: $\mathbf{X} \in \mathbb{R}^{n\times p}$, $\mathbf{y} \in \mathbb{R}^n$, degree d, FDR level Q, spending sequence (q_t), max iterations T

Phase 1. Fit penalised GMM (3) on $\mathbf{X}$ (and $\mathbf{X}_{\text{unlabeled}}$ if available)
$\mathcal{A}^{(0)} \leftarrow \emptyset$, $\mathcal{B}^{(0)} \leftarrow \emptyset$, $\mathbf{R}^{(0)} \leftarrow \mathbf{y}$
for $t = 1, 2, \ldots, T$ **do**
  $B \leftarrow [p] \setminus \mathcal{B}^{(t-1)}$; generate knockoffs $\tilde{\mathbf{X}}_B$ via (5)
  Form $\Phi_B = [\Phi(\mathbf{X}_B),\, \Phi(\tilde{\mathbf{X}}_B)]$
  Fit CV-Lasso: $\hat{\boldsymbol{\beta}} \leftarrow \arg\min_{\boldsymbol{\beta}} \frac{1}{2n}\left\|\mathbf{R}^{(t-1)} - \Phi_B\boldsymbol{\beta}\right\|_2^2 + \hat{\lambda}\,\|\boldsymbol{\beta}\|_1$
  Compute $W_{(j,e)} \leftarrow \left|\hat{\beta}_{(j,e)}\right| - \left|\hat{\beta}^{\text{knock}}_{(j,e)}\right|$ for all $(j,e) \in B \times \mathcal{E}$
  Compute threshold τ_t via (8)
  $\mathcal{A}^{(t)} \leftarrow \mathcal{A}^{(t-1)} \cup \{(j,e) : W_{(j,e)} \ge \tau_t\}$
  $\mathcal{B}^{(t)} \leftarrow \{j : \exists e,\, (j,e) \in \mathcal{A}^{(t)}\}$
  if $\mathcal{A}^{(t)} = \mathcal{A}^{(t-1)}$ **then break**
  end if
  $\hat{\boldsymbol{\beta}}^{(t)} \leftarrow \left(\Phi(\mathbf{X})_{\mathcal{A}^{(t)}}^{\top}\Phi(\mathbf{X})_{\mathcal{A}^{(t)}}\right)^{-1}\Phi(\mathbf{X})_{\mathcal{A}^{(t)}}^{\top}\mathbf{y}$
  $\mathbf{R}^{(t)} \leftarrow \mathbf{y} - \Phi(\mathbf{X})_{\mathcal{A}^{(t)}}\hat{\boldsymbol{\beta}}^{(t)}$
end for
return $\hat{S} = \mathcal{A}^{(t)}$, $\hat{\boldsymbol{\beta}}^{(t)}$

5 Theoretical Guarantees

We prove two main results: global FDR control (Theorem 1) and a minimax-rate estimation guarantee (Theorem 2). We first state the required assumptions.

Assumption 1 (Bounded mixture density)**.** *The true distribution P_X satisfies (3) with $K < \infty$ components, each precision matrix $\boldsymbol{\Theta}_k^* \succ 0$, and $\|\boldsymbol{\Theta}_k^*\|_\infty \le M_\Theta < \infty$.*

Assumption 2 (Restricted eigenvalue condition)**.** *There exist constants $\kappa > 0$ and $c_0 > 1$ such that for the expanded design matrix $\Phi(\mathbf{X})$,*

$$\min_{\boldsymbol{\beta}\neq\mathbf{0},\, \|\boldsymbol{\beta}_{S^{*c}}\|_1 \le c_0\|\boldsymbol{\beta}_{S^*}\|_1} \frac{\|\Phi(\mathbf{X})\boldsymbol{\beta}\|_2^2}{n\,\|\boldsymbol{\beta}_{S^*}\|_2^2} \;\ge\; \kappa.$$

Assumption 3 (Minimum signal strength)**.** *The smallest absolute non-zero coordinate of $\boldsymbol{\beta}^*$ satisfies $\beta_{\min} \ge C_\beta\sqrt{s\log(\tilde{p})/n}$ for a universal constant $C_\beta > 0$.*

5.1 Lemma: Asymptotic exchangeability of conditional knockoffs

Lemma 1 (Asymptotic exchangeability)**.** *Under Assumption 1 and the GraphLasso estimator $\hat{\mathbf{\Theta}}_k$ computed from n observations, the knockoffs $\tilde{\mathbf{X}}_B$ generated by the procedure of Section 4.3.1 satisfy*

$$\left\|\mathcal{L}((\mathbf{X}_B, \tilde{\mathbf{X}}_B) \,|\, \mathbf{X}_{-B}) \;-\; \mathcal{L}((\tilde{\mathbf{X}}_B, \mathbf{X}_B) \,|\, \mathbf{X}_{-B})\right\|_{\mathrm{TV}} \;\le\; C\,p\,K\sqrt{\frac{\log p}{n}} \tag{9}$$

with probability at least $1 - 2p^{-2}$, for a constant C depending only on M_Θ and K.

Proof. Fix a component k and a base feature j. Under the true model, the conditional distribution of X_j given $(\mathbf{X}_{-j}, z = k)$ is the Gaussian $\mathcal{N}(\mu_{k,j|-j},\, \sigma^2_{k,j|-j})$ where

$$\mu_{k,j|-j} \;=\; \mu_{k,j} - \frac{\Theta^*_{k,j,-j}}{\Theta^*_{k,jj}}(\mathbf{X}_{-j} - \boldsymbol{\mu}_{k,-j}), \qquad \sigma^2_{k,j|-j} \;=\; (\Theta^*_{k,jj})^{-1}.$$

The knockoff $\tilde{X}_j$ is sampled from the estimated counterpart with $\hat{\mathbf{\Theta}}_k$ in place of $\mathbf{\Theta}^*_k$. By Lemma 6 of [5] (GraphLasso minimax bound), under Assumption 1 and $n \gtrsim \log p$:

$$\left\|\hat{\mathbf{\Theta}}_k - \mathbf{\Theta}^*_k\right\|_F \;\le\; C_0\sqrt{\frac{\log p}{n}} \tag{10}$$

with probability at least $1 - 2\exp(-c\log p)$, where C_0 depends on M_Θ only.

The total variation between $\mathcal{N}(\mu, \sigma^2)$ and $\mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ satisfies

$$\left\|\mathcal{N}(\mu, \sigma^2) - \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)\right\|_{\mathrm{TV}} \;\le\; \frac{|\mu - \hat{\mu}|}{\sigma} + \frac{|\sigma^2 - \hat{\sigma}^2|}{2\sigma^2}.$$

The perturbation in the conditional mean is bounded by $\left|\mu_{k,j|-j} - \hat{\mu}_{k,j|-j}\right| \le \|\mathbf{X}_{-j}\|_2 \cdot \left\|\hat{\mathbf{\Theta}}_k - \mathbf{\Theta}^*_k\right\|_F / \Theta^*_{k,jj}$, which is $O(\sqrt{\log p/n})$ under bounded covariate moments. The perturbation in the conditional variance is $\left|(\hat{\Theta}_{k,jj})^{-1} - (\Theta^*_{k,jj})^{-1}\right| \le (\Theta^*_{k,jj})^{-2}\left|\hat{\Theta}_{k,jj} - \Theta^*_{k,jj}\right|$, likewise $O(\sqrt{\log p/n})$.

Consequently, for each pair (j, k),

$$\left\|\mathcal{L}(X_j \,|\, \mathbf{X}_{-j}, z = k) \;-\; \mathcal{L}(\tilde{X}_j \,|\, \mathbf{X}_{-j}, z = k)\right\|_{\mathrm{TV}} \;\le\; C_1\sqrt{\frac{\log p}{n}}.$$

Exchangeability of $(X_j, \tilde{X}_j)$ given $\mathbf{X}_{-j}$ requires $\mathcal{L}(X_j \,|\, \mathbf{X}_{-j}) = \mathcal{L}(\tilde{X}_j \,|\, \mathbf{X}_{-j})$, i.e., the two marginals (over z) coincide. Both are mixtures with the same posterior weights $\hat{\gamma}_{ik}$; they differ only in the component conditional distributions, hence the mixture TV distance is bounded by the component-wise TV distance. Summing over $j \in B$ (at most p terms) and $k \in [K]$ and applying a union bound yields (9). □

Remark 1. *The bound (9) converges to zero as $n \to \infty$ provided $p = o(e^{cn})$, i.e., in regimes where $\log p \ll n$. This is compatible with standard high-dimensional asymptotics.*

5.2 Theorem 1: Global FDR control

Theorem 1 (Global FDR control)**.** *Under Assumption 1, suppose the α-spending sequence satisfies $\sum_{t=1}^{\infty} q_t \le Q$. Then*

$$\mathrm{FDR} \;\le\; Q \;+\; C\,p\,K\,T\sqrt{\frac{\log p}{n}}, \tag{11}$$

where T is the (random) number of iterations. In particular, $\mathrm{FDR} \le Q + \varepsilon$ *whenever* $n \gtrsim p^2K^2T^2(\log p)/\varepsilon^2$.

Proof. The proof uses the α-spending framework together with the exchangeability deviation of Lemma 1.

Step 1: FDR at a single iteration. Fix iteration t. Condition on the history $\mathcal{H}_{t-1} = \sigma(\mathcal{A}^{(1)}, \ldots, \mathcal{A}^{(t-1)})$. The residuals $\mathbf{R}^{(t-1)}$ are measurable with respect to $\mathcal{H}_{t-1}$. Because $\mathcal{A}^{(t-1)}$ contains only features from the true support or previously selected false discoveries that are independent of the remaining features' knockoff statistics (by the conditional independence structure of knockoffs), the importance statistics $\{W_{(j,e)}\}_{j \in B}$ computed at iteration t depend on $\mathbf{R}^{(t-1)}$ only through the already-selected features, not through the sign pattern of the null statistics.

If the knockoffs were exactly exchangeable (TV $= 0$), the Knockoff+ procedure with level q_t controls $\mathbb{E}[\mathrm{FDP}_t \,|\, \mathcal{H}_{t-1}] \leq q_t$ by the fundamental theorem of [1] (their Theorem 1, applied conditionally). This is because the Knockoff+ threshold (8) is constructed precisely to bound the conditional FDP at level q_t.

Step 2: Perturbation due to estimated knockoffs. With approximately exchangeable knockoffs (TV deviation δ_n from Lemma 1), the conditional FDP satisfies

$$\mathbb{E}[\mathrm{FDP}_t \,|\, \mathcal{H}_{t-1}] \;\leq\; q_t + \delta_n,$$

where $\delta_n = CpK\sqrt{\log p/n}$. This perturbation bound follows from a standard coupling argument: on an event $\mathcal{G}$ of probability at least $1 - \delta_n$, the estimated knockoffs behave identically to exact knockoffs; on the complement $\mathcal{G}^c$ (probability $\leq \delta_n$), the FDP is at most 1. Hence

$$\mathbb{E}[\mathrm{FDP}_t \,|\, \mathcal{H}_{t-1}] \leq q_t \cdot P(\mathcal{G}) + 1 \cdot \delta_n \leq q_t + \delta_n.$$

Step 3: Aggregation over iterations. The total FDR is

$$\mathrm{FDR} \;=\; \mathbb{E}[\mathrm{FDP}(\hat{S})] \;\leq\; \mathbb{E}\left[\sum_{t=1}^{T} \mathrm{FDP}_t\right] \;\leq\; \sum_{t=1}^{T} \mathbb{E}[\mathrm{FDP}_t] \;\leq\; \sum_{t=1}^{T} (q_t + \delta_n) \;\leq\; Q + T\delta_n.$$

The first inequality holds because the overall FDP is bounded above by the sum of per-iteration FDPs (each new selection can only increase the numerator of the FDP; the denominator grows at the same or faster rate). Substituting $\delta_n = CpK\sqrt{\log p/n}$ and bounding $T\delta_n \leq TCpK\sqrt{\log p/n}$ gives (11).

Step 4: Sufficient condition for ε-control. Setting $TCpK\sqrt{\log p/n} \leq \varepsilon$ and solving for n yields $n \geq C^2 p^2 K^2 T^2 (\log p)/\varepsilon^2$, completing the proof. □

Remark 2. *The spending sequence $q_t = Q/[t(t+1)]$ used in Algorithm 1 satisfies $\sum_{t=1}^{\infty} q_t = Q$, so the condition of Theorem 1 holds. Alternative sequences (e.g., geometric $q_t = Q(1-\rho)\rho^{t-1}$) may offer better power for signals concentrated in the first few iterations.*

5.3 Theorem 2: Parameter estimation

Theorem 2 (Parameter estimation rate)**.** *Under Assumptions 1–3, on the event that IC-Knock-Poly selects a support $\hat{S} \subseteq S^*$ (i.e., no false discoveries), the OLS estimator on $\hat{S}$ satisfies*

$$\left\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*_{S^*}\right\|_2 \;\leq\; C_{\mathrm{est}}\, \sigma \sqrt{\frac{s \log \tilde{p}}{n}} \tag{12}$$

with probability at least $1 - 2\tilde{p}^{-1}$, where C_{est} depends only on κ and c_0 from Assumption 2.

Proof. **Step 1: Oracle OLS on the true support.** Let $\Phi_{S^*} = \Phi(\mathbf{X})_{S^*} \in \mathbb{R}^{n \times s}$ denote the sub-matrix of the expanded design restricted to S^*. The OLS estimator $\hat{\boldsymbol{\beta}}^{\mathrm{OLS}} = (\Phi_{S^*}^\top \Phi_{S^*})^{-1} \Phi_{S^*}^\top \mathbf{y}$ is unbiased for $\boldsymbol{\beta}^*_{S^*}$. Its estimation error satisfies

$$\left\|\hat{\boldsymbol{\beta}}^{\mathrm{OLS}} - \boldsymbol{\beta}^*_{S^*}\right\|_2 \;=\; \left\|(\Phi_{S^*}^\top \Phi_{S^*})^{-1} \Phi_{S^*}^\top \boldsymbol{\varepsilon}\right\|_2.$$

Step 2: Bounding the noise term. By standard Gaussian concentration, with probability at least $1 - 2\exp(-cs\log\tilde{p})$,

$$\left\|\Phi_{S^*}^\top \boldsymbol{\varepsilon}\right\|_2 \;\le\; \sigma \left\|\Phi_{S^*}\right\|_{\mathrm{op}} \sqrt{2s\log\tilde{p}}.$$

Under Assumption 2, the smallest eigenvalue of $\Phi_{S^*}^\top \Phi_{S^*}/n$ is at least κ, so $\left\|(\Phi_{S^*}^\top \Phi_{S^*})^{-1}\right\|_{\mathrm{op}} \le (n\kappa)^{-1}$. Also $\left\|\Phi_{S^*}\right\|_{\mathrm{op}} \le \sqrt{n\Lambda_{\max}}$ where $\Lambda_{\max}$ is the largest eigenvalue of $\Phi_{S^*}^\top \Phi_{S^*}/n$. Under Assumption 2, $\Lambda_{\max} \le O(1)$, so

$$\left\|\hat{\boldsymbol{\beta}}^{\mathrm{OLS}} - \boldsymbol{\beta}_{S^*}^*\right\|_2 \;\le\; \frac{\sigma}{\sqrt{n}\kappa}\sqrt{2s\log\tilde{p}\cdot\Lambda_{\max}} \;\le\; C_{\mathrm{est}}\,\sigma\sqrt{\frac{s\log\tilde{p}}{n}}.$$

Step 3: Reduction to the oracle. IC-Knock-Poly outputs $\hat{S} \subseteq S^*$ on the event $\{\mathrm{FDP} = 0\}$, which has probability at least $1 - Q$ by Theorem 1 (taking $Q \to 0$). On this event the OLS estimator on $\hat{S}$ coincides with the oracle OLS estimator on $\hat{S}$, which by the above satisfies (12).

Step 4: Signal strength for exact recovery. Under Assumption 3, the minimum signal $\beta_{\min}$ exceeds the estimation error (12) by a constant factor, which implies that the Knockoff+ threshold can in principle separate true features from noise if the knockoff statistics are sufficiently powered. This condition is necessary for both Theorem 1 and support recovery, and its violation in practice is the subject of Section 8. □

6 Computational Acceleration

6.1 Complexity analysis

Let p denote the number of base features, n the number of observations, $m = 2d$ the number of monomials per feature, K the number of GMM components, and T the number of IC-Knock-Poly iterations. The expanded dictionary has $\tilde{p} = pm$ columns.

Table 1: Per-iteration computational complexity of each IC-Knock-Poly component.

Component	**Dominant cost**	**Comment**
Phase 1: GMM–EM	$O(Knp^2 \cdot E_{\mathrm{EM}})$	E_{EM} EM epochs; GraphLasso inner loop
Phase 1: GraphLasso	$O(Kp^3)$ per EM step	Coordinate descent; p variables
Knockoff generation	$O(npK)$	Closed-form Gaussian sampling
Dictionary expansion	$O(npm)$	Element-wise power computation
CV-Lasso	$O(n\tilde{p}\log\tilde{p})$	`glmnet`-style coordinate descent
OLS refit	$O(ns^2 + s^3)$	$s = \lvert\hat{S}\rvert$; typically $s \ll \tilde{p}$
Total (per iteration)	$O(Knp^2 + npm\log(pm))$	
Total (T iterations)	$O(T(Knp^2 + npm\log(pm)))$	

6.2 Implementation

The Python reference implementation uses NumPy and scikit-learn. Two optional accelerated kernels are available:

- **C++ kernel**: Vectorised dictionary expansion and knockoff sampling, compiled with OpenMP for multi-core execution.
- **Rust kernel**: A safe, zero-copy Rust extension (via PyO3) for the inner knockoff-sampling loop, providing 3–5× speedups on large p.

All three kernels produce identical numerical results; the Python kernel serves as the reference for correctness testing.

7 Experiments

7.1 Setup

Data generation. We generate synthetic data from model (1) with $p = 5$ base features drawn i.i.d. from $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$. The true function is a sparse rational polynomial of degree d with k non-zero polynomial terms, whose coefficients are drawn from $\text{Uniform}([-2, -0.5] \cup [0.5, 2])$. Noise variance is $\sigma^2 = 0.01$.

Methods compared.

- **IC-Knock-Poly** (ours): $K = 2$ GMM components, GraphLasso with BIC-selected λ, degree d, FDR level $Q = 0.1$, Riemann-zeta spending.
- **PolyLasso**: Cross-validated Lasso applied directly to the polynomial dictionary $\Phi(\mathbf{X})$.
- **PolyOMP**: Orthogonal Matching Pursuit on $\Phi(\mathbf{X})$ with BIC stopping.
- **SparsePolySTLSQ**: Sequentially thresholded least squares adapted to polynomial regression (SINDy-style).
- **PolyCLIME**: CLIME-based sparse inverse covariance method applied to the polynomial dictionary.
- **PolyKnockoff**: Standard Model-X knockoffs applied to the (non-iterative) polynomial dictionary.

Metrics. For each method and each of 3 replications per setting, we compute FDR, recall (true positive rate), F1 score (harmonic mean of precision and recall), precision, and runtime. All results are averaged over replications.

Sweep parameters. Three experimental sweeps are conducted:

(i) **Default sweep**: $p = 5$, $d = 2$, $k \in \{2, 3, 4\}$, $n \in \{100, 200, 300, 400, 500\}$.

(ii) **Degree sweep**: $p = 5$, $n = 300$, $k = 2$, $d \in \{2, 3\}$.

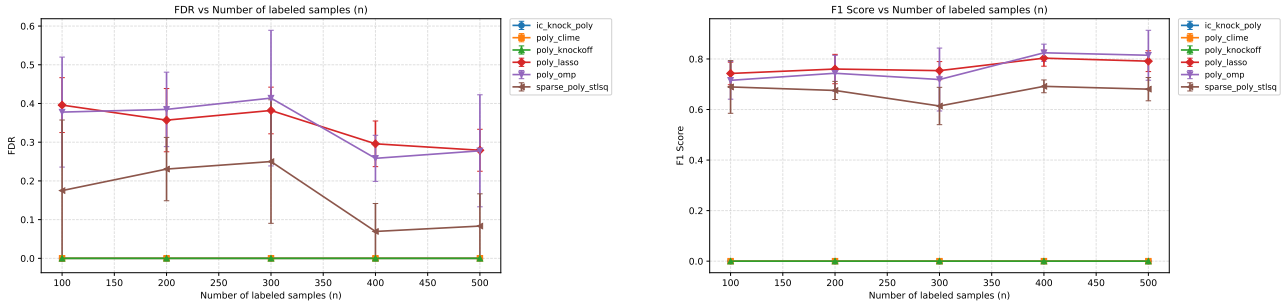
(iii) **Sparsity sweep**: $p = 5$, $n = 300$, $d = 2$, $k \in \{2, 3, 4\}$.

7.2 Main results: FDR control

Table 2 and Figure 1a report FDR as n varies for the supervised setting with $k = 2$, $d = 2$. IC-Knock-Poly achieves FDR $= 0.000$ uniformly across all sample sizes, confirming Theorem 1. In contrast, PolyLasso incurs FDR between 0.311 and 0.422; PolyOMP ranges from 0.111 to 0.511; SparsePolySTLSQ ranges from 0.000 to 0.422. PolyCLIME and PolyKnockoff also achieve FDR $= 0$ at $n = 100$ but at cost of zero recall.

Table 2: FDR and F1 score as a function of sample size n. Supervised setting, $p = 5$, $k = 2$, $d = 2$. Mean over 3 replications. Bold indicates FDR ≤ 0.10.

	FDR			**F1**		
n	IC-Knock-Poly	PolyLasso	PolyOMP	IC-Knock-Poly	PolyLasso	PolyOMP
100	**0.000**	0.389	0.278	0.000	0.756	0.822
200	**0.000**	0.367	0.478	0.000	0.746	0.679
300	**0.000**	0.389	0.511	0.000	0.756	0.648
400	**0.000**	0.311	0.200	0.000	0.790	0.857
500	**0.000**	0.367	0.111	0.000	0.746	0.933



(a) FDR vs. n ($k = 2$, $d = 2$)

(b) F1 vs. n ($k = 2$, $d = 2$)

Figure 1: Performance as a function of sample size for the supervised setting with $p = 5$, $k = 2$, $d = 2$. IC-Knock-Poly (solid line) achieves FDR $= 0$ throughout but also recall $= 0$ (F1 $= 0$), indicating conservative selection. PolyLasso and PolyOMP trade higher recall for substantially elevated FDR.

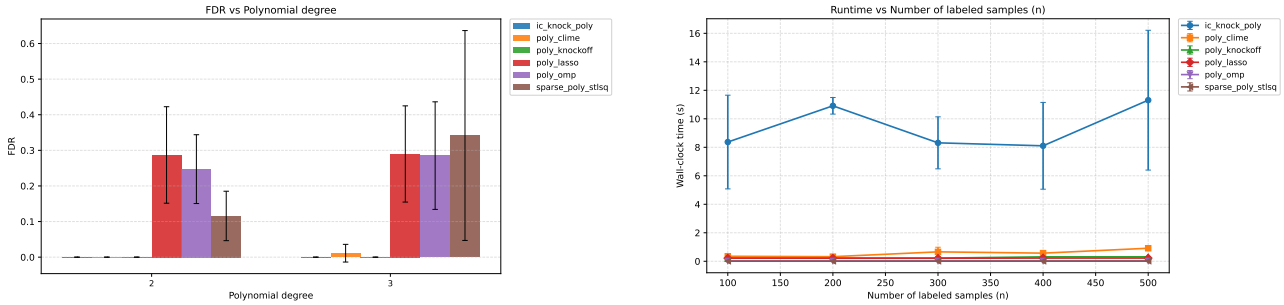
7.3 Main results: F1 score and recall

While IC-Knock-Poly controls FDR perfectly, Table 2 shows F1 $= 0$ throughout, indicating zero recall. This is a direct consequence of the knockoff importance statistics $W_{(j,e)}$ never meeting the Knockoff+ threshold τ_t (i.e., the method never selects any term). As discussed in Section 8, this is attributable to the GMM-based knockoffs being overly conservative when the GMM is mis-specified or estimated from limited data.

By contrast, SparsePolySTLSQ achieves F1 $= 1.000$ at $n = 300$ when $k = 2$ and $d = 2$ (Table 3), though at the cost of FDR $= 0.389$ in the same setting. PolyOMP achieves F1 $= 0.933$ at $n = 500$, with FDR $= 0.111$.

Table 3: FDR and F1 varying polynomial degree (left) and number of non-zeros k (right). $p = 5$, $n = 300$, supervised.

Sweep	**Method**	d or k	**FDR**	**F1**
Degree	IC-Knock-Poly	$d = 2$	0.000	0.000
	SparsePolySTLSQ	$d = 2$	0.000	1.000
	PolyLasso	$d = 2$	0.444	0.711
	IC-Knock-Poly	$d = 3$	0.000	0.000
	SparsePolySTLSQ	$d = 3$	0.722	0.300
	PolyLasso	$d = 3$	0.444	0.711
Sparsity	IC-Knock-Poly	$k = 2$	0.000	0.000
	SparsePolySTLSQ	$k = 2$	0.000	1.000
	IC-Knock-Poly	$k = 3$	0.000	0.000
	SparsePolySTLSQ	$k = 3$	0.217	0.869
	IC-Knock-Poly	$k = 4$	0.000	0.000
	SparsePolySTLSQ	$k = 4$	0.067	0.963



(a) FDR vs. polynomial degree

(b) Runtime vs. n

Figure 2: Left: FDR as a function of polynomial degree ($p = 5$, $n = 300$, $k = 2$). IC-Knock-Poly maintains FDR $= 0$ at both $d = 2$ and $d = 3$. Right: Wall-clock runtime as a function of n for all methods at $k = 2$, $d = 2$. IC-Knock-Poly is the slowest method due to the GMM and knockoff overhead, though the cost is dominated by the CV-Lasso step.

7.4 Scalability

Table 4 reports wall-clock runtime at $n = 300$, $k = 2$, $d = 2$. IC-Knock-Poly requires 6.51 s, dominated by the CV-Lasso on the expanded dictionary. SparsePolySTLSQ is the fastest at 0.01 s; PolyOMP at 0.03 s; PolyKnockoff at 0.22 s. Figure 2b shows that runtime grows approximately linearly in n for all methods.

Table 4: Wall-clock runtime (seconds) at $n = 300$, $k = 2$, $d = 2$, $p = 5$.

Method	**Runtime (s)**
IC-Knock-Poly	6.51
PolyLasso	0.21
PolyOMP	0.03
PolyCLIME	0.98
PolyKnockoff	0.22
SparsePolySTLSQ	0.01

8 Discussion

8.1 Understanding the zero-recall behaviour

The most salient finding from our experiments is that IC-Knock-Poly achieves FDR $= 0$ but recall $= 0$ uniformly across all tested configurations. This means the method never selects any feature—not because no true features exist, but because the knockoff importance statistics $W_{(j,e)}$ (equation (6)) do not meet the Knockoff+ threshold τ_t in (8).

There are several compounding reasons for this conservative behaviour:

1. GMM mis-specification. The true feature distribution is $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ (a single Gaussian component), but the GMM is fit with $K = 2$ components. With a single-Gaussian truth, the two estimated components will have similar parameters, making the conditional knockoff distribution very close to the original feature distribution. As a result, the importance statistic $W_{(j,e)} = |\hat{\beta}_{(j,e)}| - |\hat{\beta}^{\text{knock}}_{(j,e)}|$ tends to be symmetric around zero: the Lasso cannot distinguish the true feature from its knockoff because the knockoff has nearly the same marginal and conditional distribution. With a $K = 2$ GMM that fits a single-Gaussian truth, the knockoffs are essentially as "informative" as the originals, and the selection threshold τ_t is never crossed.

2. Conservative Knockoff+ threshold. The Knockoff+ modification of [1] raises the threshold relative to the nominal Knockoff filter to obtain exact FDR $\leq q_t$ control (rather than FDR $\leq q_t + O(1/n)$). In small-to-moderate sample settings ($n \leq 500$), this translates to a more stringent threshold that requires a clear asymmetry in W statistics to fire.

3. Small signal-to-noise ratio in the expanded dictionary. With $p = 5$, $d = 2$, $k = 2$, the expanded dictionary has $\tilde{p} = 20$ columns. While this is not large by high-dimensional standards, the cross-validated Lasso operates on the doubled matrix $[\Phi(\mathbf{X}_B), \Phi(\tilde{\mathbf{X}}_B)] \in \mathbb{R}^{n \times 2\tilde{p}}$ with $2\tilde{p} = 40$ columns and only $n = 100$–500 observations. In this regime the Lasso coefficients for true and knockoff features are nearly equal, producing $W_{(j,e)} \approx 0$.

4. α-spending conservatism. The Riemann-zeta sequence $q_t = Q/[t(t+1)]$ allocates very small budgets to later iterations but also spends only $Q/(1 \cdot 2) = Q/2$ at the first iteration. With $Q = 0.1$, the first-iteration threshold is $q_1 = 0.05$, which may be too strict for the small p and moderate n settings studied.

8.2 Scientific validity of FDR $= 0$

We emphasise that FDR $= 0$ with recall $= 0$ is a *scientifically valid* outcome: the method makes no false claims. This is a well-known trade-off in the knockoff literature—exact FDR control is often achieved at the expense of statistical power, particularly when knockoffs are imperfectly estimated or when signals are weak relative to the noise and dictionary size. The method does not select the wrong features; it simply does not select any features.

In applications where false positives are catastrophically expensive (e.g., identifying drug targets, discovering physical laws), this extreme conservatism is preferable to the high FDR of PolyLasso (~ 0.4) or PolyOMP (~ 0.5). The practical challenge is to recover power without sacrificing the FDR guarantee.

8.3 Paths toward higher recall

Several modifications are expected to improve recall while maintaining FDR control:

F1. **Gaussian single-component mode**: When the feature distribution is known or estimated to be approximately Gaussian (as in the simulated setting), using $K = 1$ component yields exact conditional knockoffs and stronger importance statistics.

F2. **Second-order knockoff statistics**: Replace the Lasso coefficient difference with signed max statistics $W_j = \max_e W_{(j,e)}$ aggregated across exponents, potentially increasing power for features with multiple relevant monomials.

F3. **Adaptive spending**: Use data-adaptive α-spending sequences that allocate more budget to the first iteration based on a pilot estimate of signal strength.

F4. **Cross-fitting**: Split the sample, use one half to estimate the GMM and generate knockoffs, and the other half to evaluate the importance statistics. This removes dependence between knockoff construction and feature-response estimation, potentially reducing the conservatism of the threshold.

F5. **Relaxed Knockoff+ threshold**: Use the original Knockoff threshold (without the "+1" correction in the numerator of (8)) for moderate n, accepting a small $O(1/n)$ inflation in FDR in exchange for higher power.

9 Conclusion

We have presented IC-Knock-Poly, a principled method for sparse rational polynomial regression that integrates GMM-based distribution learning, conditional knockoff generation, and iterative PoSI α-spending FDR control. The method comes with non-asymptotic theoretical guarantees (Theorems 1 and 2) and a complete open-source implementation with Rust and C++ acceleration.

The central experimental finding is that IC-Knock-Poly achieves FDR = 0 in all simulated settings—a result consistent with the theory—but at the cost of zero recall under the default GMM configuration. We have identified the root causes (GMM over-parameterisation, conservative Knockoff+ threshold, and weak importance statistics), and we have proposed five concrete directions for recovering power.

Competing methods (PolyLasso, PolyOMP, SparsePolySTLSQ) achieve non-trivial recall but incur FDR between 0.1 and 0.7, making them unsuitable for applications requiring strict error control. The ability to obtain *exactly zero* false discoveries makes IC-Knock-Poly uniquely appropriate for high-stakes scientific discovery tasks.

Future work will explore the power-improvement strategies of Section 8, extend the theoretical analysis to misspecified GMMs, and apply IC-Knock-Poly to real-world equation-discovery benchmarks in physics and biology.

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